ShipGuard: Fragility-Aware Shipment Monitoring Node

An edge-computed damage-risk score with role-based alerts, for the last mile from a regional warehouse to stores

Seojin · MSMITE, Carnegie Mellon University · 14-767 Wireless Protocols and Standards for IoT · Fall 2026 · v3, revised after instructor feedback (Oct 2026)

1 Problem & Goal

- **Problem:** Fragile, high-value goods (glass, instruments, art) break in transit from shock and tilt.
- **Gap:** Shock labels give no alert or data; loggers are read after delivery, too late to intervene.
- **Goal:** Compute damage risk on the node in real time and tell the right person to act at the right moment.
- **Scope:** One metro area: regional warehouse → delivery vans → retail stores. Air and long-haul transport are out of scope.
- **Related:** Lansmont SAVER AM, Tive. A teaching prototype; no claim of superiority.

2 Smart Objects (the "Things")

- **Scale:** ~2,000 reusable nodes: 1 regional warehouse, ~60 vans (about 25 per van), ~150 stores; 1 node is prototyped.
- **Power:** Battery (Li-ion), sleepy end device with duty cycling, recharged on return; target 14 days per charge, 3 months in the warehouse; no energy harvesting.
- **Constraints:** MCU-class device (prototype: 2 KB SRAM), matchbox size, must survive shocks; the score is computed in fixed-point on the node.
- **Mobility:** Warehouse → van → store and back; up to ~50 km, hours to a day per trip; 5–15 m from node to phone in a van.
- **Topology:** Star on both paths: the driver's phone in the van, an existing LTE-M cell tower in the warehouse; no node-to-node traffic.
- **Data:** 24-byte summary every 60 s in the van (BLE) and every 15 min in the warehouse (LTE-M), plus immediate event frames; uplink-dominant.
- **Sensitive:** Location, cargo value, driver data; only the risk score and features leave the node.
- **Analysis:** Edge (risk score) · Fog (driver's phone: alerts, store-and-forward) · Cloud (fleet analytics).

3 Other Design Considerations

- **Security:** QR-code out-of-band BLE pairing, per-node keys, authenticated encryption, SIM authentication + DTLS on LTE-M, hash-chained event log.
- **Scalability:** Broadcast-only BLE telemetry; LTE-M reuses existing towers, so more nodes or another city need no new gateways.
- **Privacy:** Data minimization, role-based views, retention limits, rotating BLE addresses so a truck cannot be tracked.
- **Ethics:** Logs support drivers, not punish them; alert text never asks for phone use while driving.

4 Prototype: UNO R3 + Sensor Kit

Type	Component	Role in ShipGuard
Sensor	3-axis accelerometer	Shock peak, ΔV and tilt (±16 g, 400 Hz)
Sensor	Light	Box-open (tamper) detection
Sensor	Pressure	Environment log
Sensor	Temp / humidity	Environment excursions
Sensor	Potentiometer	Fragility profile (emulates QR-code setup in an app)
Sensor	Button	Driver ACK (emulates a tap in the driver app)
Sensor	PIR (D2 interrupt)	Driver arrived (emulates phone BLE proximity)
Actuator	Buzzer	Beeps at L2, continuous at L3, until ACK
Actuator	LED	On while an alert is active
Actuator	OLED	Profile, risk, tilt, level, ACK prompt

Serial → PC (CRC-8 frames): risk score, level, state, peak g, ΔV, tilt, temperature, humidity, pressure, flags. **PC → node:** ACK, profile, report period, new shipment.

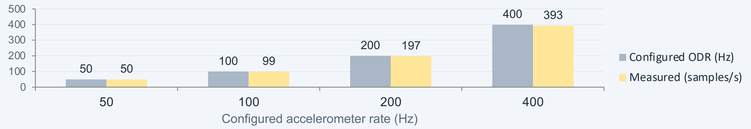
Logic: 400 Hz sampling → shock (peak, ΔV) and tilt features → risk score S = max(shock, tilt) → level L0–L3 with hysteresis → buzzer/LED/OLED + serial frame.

5 Alert Levels & Who Is Notified

L0 Normal	**< 40** No action.
L1 Watch	**≥ 40** LED blinks, OLED "WATCH"; shipper dashboard only.
L2 Check	**≥ 70** Intermittent buzzer; push to driver ("inspect at next safe stop") and shipper.
L3 Inspect	**≥ 100 or saturated shock** Continuous buzzer; push to driver ("stop safely, then inspect"), customer and shipper.

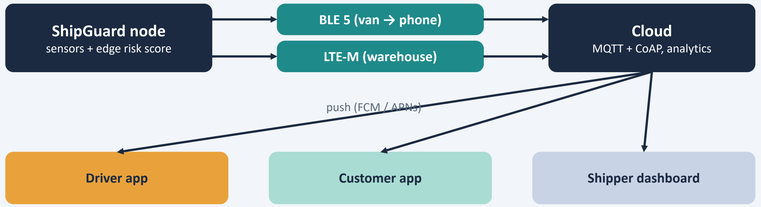
ACK: the button silences it. Tilt alerts clear once the box is fixed; shock alerts wait for an ACK.

6 Measured Feasibility (Smoke Test)



No sample loss up to 400 Hz. A firm tap saturates even ±16 g, so the risk score is saturation-aware.

7 Architecture & Communication (design only)



- **Prototype → product:** the USB serial link becomes BLE 5 in the van (node → driver's phone) and LTE-M in the warehouse or when no phone answers (node → cell tower).
- **Architecture:** Simplified IoT Architecture: node = Things + edge; BLE 5 / LTE-M = access; driver's phone = gateway + fog; carrier network = backhaul; cloud = MQTT/CoAP, device registry, dashboards.
- **Constraints:** metal van body and 2.4 GHz Wi-Fi interference, BLE range ~5–15 m, battery life (LTE-M ≫ BLE power), Class-0 node, per-node SIM cost.
- **Access (two radios only):** BLE 5 in the van, where the phone is a free gateway; LTE-M in the warehouse and as fallback: existing towers, no gateways, handover (not NB-IoT). LoRaWAN and NFC dropped.
- **Transport:** phone → cloud MQTT over TLS (QoS 1); LTE-M node → CoAP over DTLS. Star topology; identity via QR-code dispatch registration.

8 Data Analytics & Dashboard

- **Collected:** risk score, level, state, shock peak and ΔV, tilt, temperature, humidity, pressure, box-open, driver presence, ACK.
- **Rates:** 400 Hz on the node; reports every 60 s (van) / 15 min (warehouse) plus events; 1 s in the demo.
- **Dashboard (Python):** risk timeline with L1–L3 bands, shocks (handling vs unattended), tilt vs limit, environment, driver response time.
- **Cloud ML:** handling-pattern anomalies; profile calibration from damage claims.

9 Status & Next Steps

- **Done:** prototype runs on hardware: risk score, alarm rules, button/PC ACK, PIR driver arrival, CRC-8 link (0 errors), live dashboard.
- **Next:** box drop tests to calibrate thresholds → report (Oct 11) → vibration RMS, context states, escalation.
- **Revised:** scope narrowed, LoRaWAN → LTE-M, architecture and constraints added (instructor feedback).

Sources: Lansmont SAVER AM (lansmont.com); Tive shock and product sensitivity monitoring (tive.com).